



AJKD Atlas of Renal Pathology: Cortical Necrosis

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Clinical and Pathologic Features

Cortical necrosis commonly manifests clinically as a rapid loss of glomerular filtration rate, often with oliguria, granular casts, and low-level proteinuria. Cortical necrosis may occur at any age. Causes include complications of vascular anastomosis in the transplanted kidney. In the native kidney, it occurs after thrombosis of interlobular or larger arteries, massive cholesterol emboli, septic abortion, or other catastrophic obstetrical complications (including eclampsia), and in the neonate, it occurs as complications of sepsis and placental hemorrhage. Causes of native kidney cortical necrosis typically affect both kidneys. Prognosis depends on the underlying cause.

Light microscopy: There is coagulative necrosis involving all tubular segments and glomeruli. Nuclei may be pale and pyknotic, or may no longer be apparent. Thrombi may be present in vessels at the edge of the infarct.

Immunofluorescence microscopy: No specific staining.

Electron microscopy: Widespread necrosis.

Etiology/Pathogenesis

Thrombosis of interlobular or larger arteries causes cortical infarcts. Obstetric complications, particularly placental abruption and eclampsia, are common causes. Incidence is decreasing in developed countries as obstetric care is improving.

Differential Diagnosis

The extent of necrosis may not be accurately judged from a needle or even wedge biopsy. The kidney may have zones of infarct that are not accurately sampled in a biopsy. Thus, a biopsy cannot distinguish a non-viable kidney, eg, in the

transplant setting, from a kidney with a zone of necrosis.

Key Diagnostic Features

- Pale tissue with typical appearance of coagulative necrosis, affecting both tubules and glomeruli
- Pyknotic nuclei
- Simplified, flattened epithelium of proximal tubules

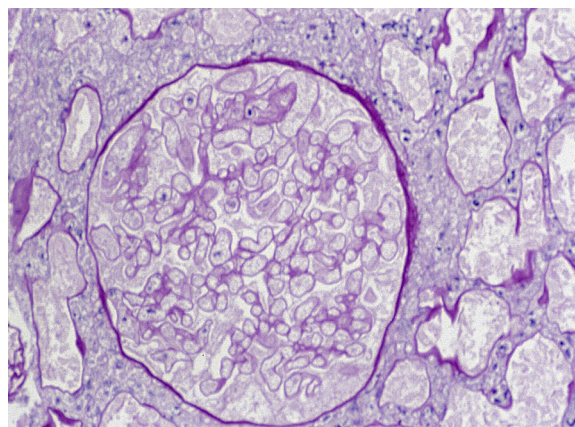


Figure 1. Cortical necrosis with coagulative necrosis of a glomerulus and tubules (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 41(1):e27.

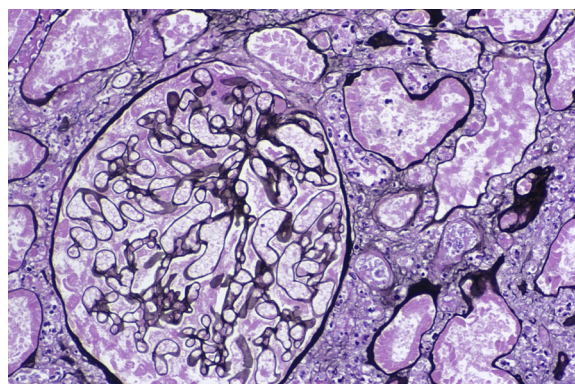


Figure 2. Cortical necrosis with tubular and glomerular necrosis. The tubular and glomerular basement membranes are still visible without viable nuclei (Jones silver stain). Reproduced with permission from *AJKD* 41(1):e27.

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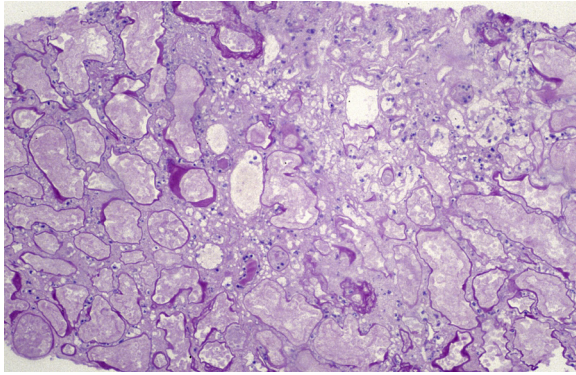


Figure 3. Cortical necrosis with frank tubular necrosis with ghost-like outlines of cells without discernible nuclei, characteristic of coagulative necrosis (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 41(1):e27.

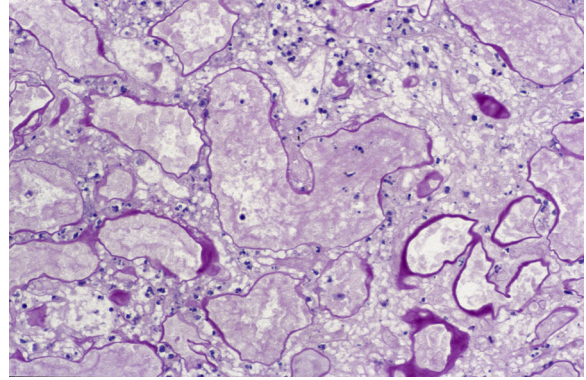


Figure 4. Cortical necrosis with frank tubular necrosis with ghost-like outlines of cells without discernible nuclei as well as pyknotic nuclei and karyorrhexis, characteristic of coagulative necrosis (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 41(1):e27.